

Active Inference

Principles and Applications

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- Focus on the principles and theories that underwrite the framework
- Mention some of the process theories that have been associated with the active inference framework
- What is active inference and why is it something worth speaking about?

Active Inference

Definition (Active Inference)

A system is conducting active inference if it adjusts its **beliefs** about the environment, as adjusting the environment so as to minimise the **surprise** of observations generated from its environment.

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Active Inference

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A system is conducting active inference if it adjusts its **beliefs** about the environment, as adjusting the environment so as to minimise the **surprise** of observations generated from its environment.

- Unify different biological processes as optimising the same underlying objective
- heterogeneity vs homogeneity
- Unification can cause problems with where to start in a discussion on active inference
- Beliefs and Surprise - strict definitions within specific contexts but there is some intuition about what they could mean but we can intuit some inferential process
- Active component



Outline

1 High Road

2 Low Road



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Active Inference

Outline

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1 High Road

2 Low Road

- Low Road - As few assumptions about the underlying systems as possible
- High Road - Build on modern theories of perception as inference



1 High Road

2 Low Road

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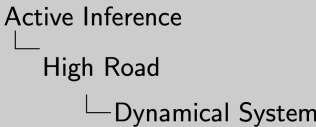
Active Inference
High Road

- 1 High Road
- 2 Low Road

Dynamical System

- Physical State Space X

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Dynamical System
• Physical State Space X

- Place the system in some mathematical framework
- Cells – Sensory states could refer to the presence of nutrients detected by receptors on the cell membrane, while internal states might include the concentration of intracellular chemicals responsible for metabolism.
- A subset of the internal states being the actions that the organism can take on the environment
- Inferential loop



Dynamical System

- Physical State Space X
- Allow for the representation $X = S \times R$

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Active Inference

└ High Road

└ Dynamical System

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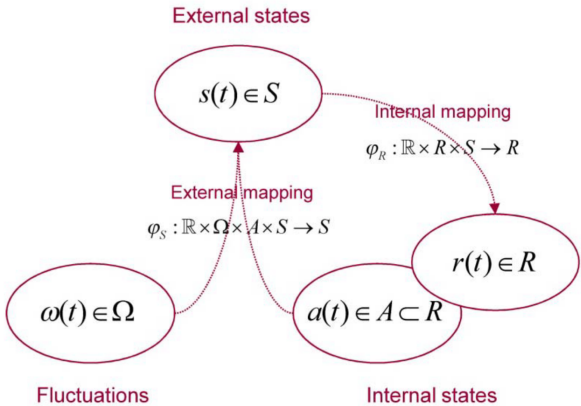


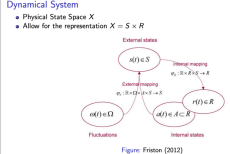
Figure: Friston (2012)

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Active Inference

└ High Road

└ Dynamical System



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Ergodic Density

Definition (Ergodic Density)

For $x \in X$. We define the ergodic density p as:

$$p(x) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \delta(x - x(t)) \, dt$$

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Active Inference

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- Use the assumption of the system's persistence through time



Definition (Active System)

We consider a biological system active if it possess an internal map φ_R^*

$$\varphi_R^* = \operatorname{argmin}_{\varphi_R} H(S) = - \int p(s) \ln p(s) \, ds$$

- Biological systems typically occupy a small subset of the possible - consider temperature of a human body

Minimisation of Surprise

Theorem (Principle of Least Action)

The internal states of an active system minimise surprise, $\mathcal{L}(s(t)) = -\ln p(s(t))$.

$$\varphi_R^*(t) = \operatorname{argmin}_r \mathcal{L}(s(t))$$

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Active Inference

High Road

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- Tie Entropy to Surprise
- Surprise here is defined in terms of the ergodic density
- Result comes from the relation between the surprise and the entropy of the sensory states through the ergodic theorem
- Principle of least action





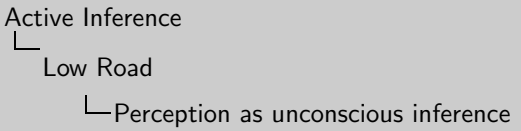
1 High Road

2 Low Road

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Active Inference
└─ Low Road

1 High Road
2 Low Road

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- What does perception as inference mean?
- Introspection may tell us that what we are perceiving is completely truth this is not replicated experimentally
- Example

Perception as unconscious inference



Perception as unconscious inference

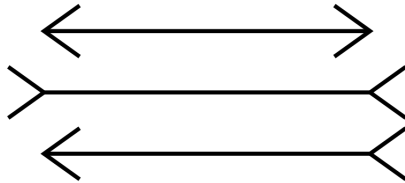


Figure: Howe and Purves (2005)

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- Active Inference
 - Low Road
 - Perception as unconscious inference

Perception as unconscious inference

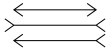


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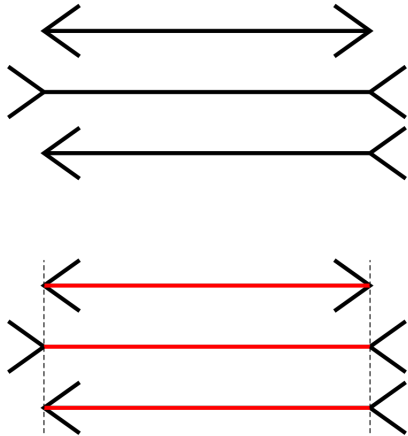
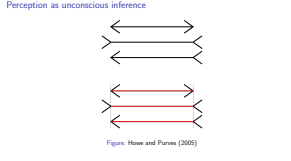


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Active Inference

- Low Road
- Perception as unconscious inference

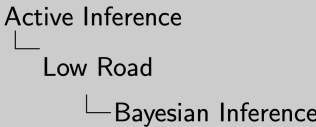


- Motivation that something else must be at play during perception - argument for combination of historical and current sensory stimuli giving rise to what we feel as perception.
- Areas that are associated with memory storage and collection are very close to areas activated during visual stimuli
- Bottom up and top down

Bayesian Inference

- Generative Model: $p(x, y)$

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Bayesian Inference

- Generative Model: $p(x, y)$

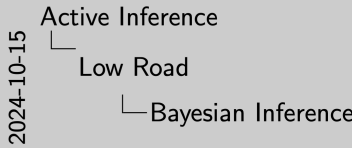
- Forward models are called generative models in the context of active inference
- Generative models are a crucial component to the notion of the brain performing inference
- Inference is about is updating probabilities distributions (beliefs) based on new information
- Interplay between historical information and new sensory input
- Intractability



Bayesian Inference

- Generative Model: $p(x, y)$
- Bayes' Rule:

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$



Bayesian Inference

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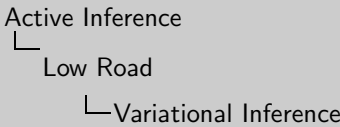
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$$p(y) = \int_{\mathcal{X}} p(x, y) \, dx \quad (1)$$

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Variational Inference

$$p(y) = \int_{\mathcal{X}} p(x, y) \, dx \quad (1)$$

- Optimality Criteria
- Set of candidates
- Optimality of Bayesian inference - constrained by the model
- Interplay of the top down and bottom up dynamics



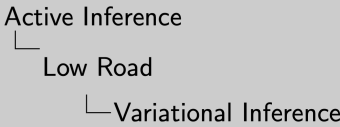
Variational Inference

$$p(y) = \int_X p(x, y) \, dx \tag{1}$$

Definition (Variational Free Energy)

$$\mathcal{F}(q, y) = \int q(x) \ln \frac{q(x)}{p(x, y)} \, dx$$

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$$\mathcal{F}(q, y) = D_{KL} [q(x) \| p(x)] - \mathbb{E}_q [\ln p(y|x)]$$

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 - Variational Inference

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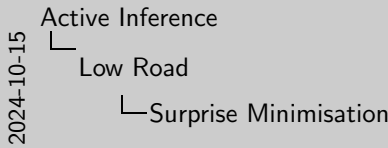
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Surprise Minimisation

$$\mathcal{F}(q, y) = D_{KL}[q(x)||p(x|y)] - \ln p(y)$$



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- Established that our brain which has been motivated as performing Bayesian inference is minimising VFE
- Our ultimate goal was to associate the the brain to active inference which is surprise minimisation
- Observation and posterior



Surprise Minimisation

$$\mathcal{F}(q, y) = D_{KL} [q(x) \| p(x|y)] - \ln p(y) \quad (2)$$
$$\geq \underbrace{-\ln p(y)}_{\text{Surprise}} \quad (3)$$

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Active Inference
└ Low Road
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Surprise Minimisation

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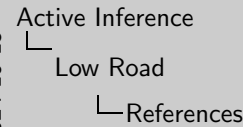
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